American Computer Science League

2020-2021 ● Contest 1: Solutions ● Classroom Division

1. Computer Number Systems

The binary representation of the decimal numbers from 22 to 32 is as follows:

22: 10110	28: 11100
23: 10111	29: 11101
24: 11000	30: 11110
25: 11001	31: 11111
26: 11010	32: 100000
27: 11011	There are 36 1's

1. 36 (C)

2. Computer Number Systems

Converting each to binary makes 101100000_2 the largest:

$101100000_2 = 101100000_2$
$535_8 = 101011101_2$
$15F_{16} = 101011111_2$

2. 101100000_2 (A)

3. Computer Number Systems

$2021_{10} = 1024 + 512 + 256 + 128 + 64 + 32 + 4 + 1$
$= 1 * 2^{10} + 1 * 2^9 + 1 * 2^8 + 1 * 2^7 + 1 * 2^6 + 1 * 2^5$
$+ 0 * 2^4 + 0 * 2^3 + 1 * 2^2 + 0 * 2^1 + 1 * 2^0$
$= 11111100101_2$
There are 8 1's.

3. 8 (C)

4. Computer Number Systems

Method 1: Subtract in octal and convert the answer to hex.
$72126_8 - 54127_8 = 15777_8 = 1101111111111_2 = 1BFF_{16}$

Method 2: Convert each number to hex and then subtract in hex.
$72126_8 - 54127_8 = 111010001010110_2 - 101100001010111_2$
$= 7456_{16} - 5857_{16} = 1BFF_{16}$

4. $1BFF_{16}$ (A)

5. Recursive Functions

Evaluate each starting at 1:

$$f(1) = 1$$
$$f(2) = f(1) + 2 = 3$$
$$f(3) = f(2) + 3 = 6$$
$$f(4) = f(3) + 4 = 10$$
$$f(5) = f(4) + 5 = 15$$
$$f(6) = f(5) + 6 = 21$$
$$f(7) = f(6) + 7 = 28$$

Note: the sequence generated is 1, 3, 6, 10, 15, 21, 28, … These are called triangular numbers - each nth term is the sum of the first n integers. The formula for that sum is $\frac{n(n+1)}{2}$. For $n = 7$, it is $\frac{7*8}{2} = 28$.

5. 28 (B)

6. Recursive Functions

Evaluate the function by working forwards, then backwards.

$$f(20) = f\left(\frac{20}{2}\right) + 1 = f(10) + 1 = 4 + 1 = 5$$
$$f(10) = f\left(\frac{10}{2}\right) + \ = f(5) + 1 = 3 + 1 = 4$$
$$f(5) = f(5 + 3) - 1 = f(8) - 1 = 4 - 1 = 3$$
$$f(8) = f\left(\frac{8}{2}\right) + 1 = f(4) + 1 = 3 + 1 = 4$$
$$f(4) = f\left(\frac{4}{2}\right) + 1 = f(2) + 1 = 2 + 1 = 3$$
$$f(2) = f\left(\frac{2}{2}\right) + 1 = f(1) + 1 = 1 + 1 = 2$$
$$f(1) = \frac{1 + 1}{2} = \frac{2}{2} = 1$$

6. 5 (D)

7. Recursive Functions

The function evaluates as follows:

$f(1) = 2 \cdot 1 + 2 = 4 \quad f(2) = 2 \cdot 2 + 2 = 6$

$f(3) = f(2) - f(1) = 2$

$f(4) = f(3) - f(2) = 2 - 6 = -4$

$f(5) = f(4) - f(3) = -4 - 2 = -6$

$f(6) = f(5) - f(4) = -6 - (-4) = -2$

$f(7) = f(6) - f(5) = -2 - (-6) = 4$

$f(8) = f(7) - f(6) = 4 - (-2) = 6$

$f(9) = f(8) - f(7) = 6 - 4 = 2$

$f(10) = f(9) - f(8) = 2 - 6 = -4$

The sequence of numbers repeats: 4, 6, 2, -4, -6, -2, …

7. -4 (B)

8. Recursive Functions

The function evaluates as follows:

$f(17) = 2 + f(17 - 2) = 2 + f(15) = 2 + 11 = 13$

$f(15) = 2 + f(15 - 2) = 2 + f(13) = 2 + 9 = 11$

$f(13) = 2 + f(13 - 2) = 2 + f(11) = 2 + 7 = 9$

$f(11) = 2 + f(11 - 2) = 2 + f(9) = 2 + 5 = 7$

$f(9) = 2 + f(9 - 2) = 2 + f(7) = 2 + 3 = 5$

$f(7) = 7 - 4 = 3$

Therefore, $f(f(17)) = f(13) = 9$

8. 9 (A)

9. What Does This Program Do? (Branching)

The following table can be used to trace the program:

a	b	c
24	2	3
24	6	3
6	6	3
6	8	3
6	8	2

So a + b / c ^ 2 = 6 + 8 / 2 ^2 = 6 + 8 / 4 = 6 + 2 = 8

9. 8 (A)

10. What Does This Program Do? (Branching)

The program can be traced using the following table:

a	b	c	d
5	2	25	4
5	6	25	4
5	6	10	4
5	6	10	1
5	6	10	1
5	6	10	5

There are 7 unique values of the variables: 1, 2, 4, 5, 6, 10, 25.

10. 7 (D)